

Retail Store Sales Analysis — Analysis Report

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Executive Summary

This report summarizes the dataset quality, exploratory findings, and model performance for project "Retail Store Sales Analysis".

Dataset Overview

File	retail_sales_dataset.csv
Rows	3000
Columns	14
Missing Values	29
Duplicate Rows	0
Cleaned	True

Modeling & Evaluation

Model Type	gradient_boosting
Problem Type	regression
Target Column	sales_amount

Metrics

Metric	Value
mae	48.2551
mse	9857.2529
rmse	99.2837
r2	0.9838
mape	0.1481

actual_vs_predicted	[[{'actual': 1406.83, 'predicted': 1360.2839}, {'actual': 21.58, 'predicted': 10.7719}, {'actual': 166.96, 'predicted': 167.0487}, {'actual': 81.68, 'predicted': 87.7463}, {'actual': 93.14, 'predicted': 110.1106}, {'actual': 658.41, 'predicted': 784.5566}, {'actual': 330.05, 'predicted': 258.9392}, {'actual': 128.48, 'predicted': 265.3418}, {'actual': 169.33, 'predicted': 222.9421}, {'actual': 32.73, 'predicted': 33.052}, {'actual': 332.1, 'predicted': 309.6851}, {'actual': 93.58, 'predicted': 81.2721}, {'actual': 1075.78, 'predicted': 989.5971}, {'actual': 163.58, 'predicted': 157.2237}, {'actual': 1690.33, 'predicted': 1636.5222}, {'actual': 1472.77, 'predicted': 1447.1805}, {'actual': 1890.78, 'predicted': 1429.6869}, {'actual': 659.49, 'predicted': 607.3261}, {'actual': 27.11, 'predicted': 37.6552}, {'actual': 814.17, 'predicted': 733.6198}, {'actual': 1473.26, 'predicted': 1636.6709}, {'actual': 77.2, 'predicted': 102.8104}, {'actual': 311.39, 'predicted': 338.9172}, {'actual': 1338.82, 'predicted': 1234.2298}, {'actual': 933.54, 'predicted': 1205.6059}, {'actual': 84.42, 'predicted': 157.475}, {'actual': 33.99, 'predicted': 35.2828}, {'actual': 234.26, 'predicted': 232.1441}, {'actual': 138.04, 'predicted': 161.6576}, {'actual': 151.0, 'predicted': 140.3772}]]
residuals_sample	[46.5461, 10.8081, -0.0887, -6.0663, -16.9706, -126.1466, 71.1108, -136.8618, -53.6121, -0.322, 22.4149, 12.3079, 86.1829, 6.3563, 53.8078, 25.5895, 461.0931, 52.1639, -10.5452, 80.5502, -163.4109, -25.6104, -27.5272, 104.5902, -272.0659, -73.055, -1.2928, 2.1159, -23.6176, 10.6228]
cv_scores	[0.980918491626912, 0.9749943104055406, 0.9774564922924323, 0.9798875564350946, 0.9884735357563104]
cv_mean	0.9803
baseline	{'baseline_score': -0.0031, 'baseline_metric': 'r2', 'model_score': 0.9838, 'improvement': 0.9869, 'cv_std': 0.0046, 'clearly_beats_baseline': True}
leakage_warnings	[]
post_outcome_excluded	[]
post_outcome_included	[]
is_weak	False
weak_reason	None

Feature Importance

Feature	Importance
unit_price	0.3644
profit	0.3203

quantity	0.163
product_category_Electronics	0.1504
discount_percent	0.0005
product_category_Groceries	0.0004
customer_rating	0.0003
customer_segment_Regular	0.0002
order_date_dayofweek	0.0001
sales_channel_Mobile App	0.0001
store_city_Lahore	0.0001
order_date_month	0.0001
promotion_applied_No	0.0001
order_date_year	0.0001
store_city_Islamabad	0.0
customer_segment_Premium	0.0
product_category_Home & Kitchen	0.0
store_city_Karachi	0.0
promotion_applied_Yes	0.0
product_category_Beauty	0.0

AI Business Insight AI-generated insight

AI service is currently unavailable. This report contains verified analytical results only; AI-generated narrative could not be produced.

Limitations

Model predictions are estimates based on historical data and may contain uncertainty. AI-generated text explains verified analytical results but does not independently calculate metrics. Always validate significant business decisions with domain expertise.